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Determination of the biochemical methane potential of food waste after aerobic storage and aeration pretreatment

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Received May 25 2022; Revised June 21 2022; Accepted July 07 2022;
View online August 01, 2022 at Wiley Online Library (wileyonlinelibrary.com);
DOI: 10.1002/bbb.2414; *Biofuels, Bioprod. Bioref.* 16:1733–1743 (2022)

Abstract: Food waste represents a relevant fraction of municipal solid organic waste, and anaerobic digestion can be an eco-friendly alternative for sustainable management in a circular economy framework. One of the challenges to the implementation of anaerobic digestion is the availability of easily degradable compounds in the substrate, which can be solved by the application of pretreatment to increase methane production. In this study, the influence of aerobic storage time and forced continuous aeration pretreatment on the biochemical methane potential of food waste was evaluated. The results demonstrate an increase in the methane production concerning the total volatile solids (TVS) of food waste stored for 7 days (425 NmL CH_4 g^{-1} TVS) compared with fresh samples without pretreatment (375 NmL CH_4 g^{-1} TVS). The adoption of forced and continuous aeration pretreatment on food waste for 4 days produced 456 NmL CH_4 g^{-1} TVS for the leachate, 1.22-fold higher than that for the food waste without storage. In conclusion, the application of aeration pretreatment previous to anaerobic digestion can be an alternative to increase the methane potential from food waste. © 2022 Society of Chemical Industry and John Wiley & Sons, Ltd.

Key words: anaerobic digestion; biogas; bioenergy; VDI 4630; Buswell model; energy; circular economy

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Introduction

Nowadays, it is estimated that about 33.3% of all food produced globally for human consumption is lost in the supply chain, including agricultural production, processing, distribution, consumption and postharvest activities, which is equivalent to 1.3 billion tonnes per year.[1] Therefore, there is a growing interest in the management of food waste, especially owing to the environmental problems caused by the improper disposal of solid waste.[2] Eco-friendly technologies for waste management have been studied as an alternative for bioenergy recovery, considering the circular economy concept.[3,4] The circular economy is a conceptual model widely used in waste management, and organic wastes are feedstocks that are well suited to playing a role in sustainable development based on the bioeconomy.[5,6] Moreover, the bioenergy recovered from organic wastes can be a substitute for fossil fuels, reducing environmental impacts and contributing to the energy matrix decarbonization.[7,8]

Anaerobic digestion (AD) technology allows the treatment of the organic fraction of municipal solid waste (OFMSW) and the production of methane-rich biogas and fertilizer for agricultural use.[9,10] The large-scale implementation of AD of the OFMSW, of which food waste (FW) represents a significant part, is a challenge for dissemination of the technology, especially owing to the low biodegradation associated with the presence of lignocellulosic compounds in the solid phase.[11,12] Regarding the process stability and efficiency, the use of FW as a substrate in AD reactors shows some problems associated with the accumulation of volatile fatty acids and the inhibition of methane production,[13] particularly in the start-up of batch garage-type reactors, where unbalanced conditions occur owing to the rapid degradation of easily degradable compounds.[14]

Hence, the pretreatment of the FW previous to AD is a significant factor in increasing methane production.[15] The physicochemical pretreatment of FW has been conducted to increase hydrolysis and facilitate the subsequent methanization.[16,17] Pretreatments with microwave, ultrasound, ozonization, mechanical fragmentation, thermal treatment and enzymatic hydrolysis are expensive for FW and decrease the energy potential of AD.[18,19] One of the alternative pretreatments to AD is the simple storage of FW under appropriate aerobic conditions, which can make the biomass suitable for methanization.[20] However, the influence of the storage period on the subsequent AD of food waste is still unclear.[21] The pretreatments can provide higher biogas yields and faster establishment of methanogenic microbiota.[22] Another pretreatment identified as a facilitator for further methanization is continuous aeration treatment.[23] Compared with various types of mechanical, thermal or chemical treatments, it efficiently reduces biodegradable compounds, improves the production of hydrolytic enzymes and has low energy consumption.[24]

The main advantages of incorporating aeration in hydrolytic-acidogenic reactors are the solubilization of organic substrates and improvement of the hydrolysis phase, which represents a positive effect on the methanogenic phase.[25] In the specific case of garage-type reactors, this can be even more important owing to the high concentration of solids in the substrate to be digested. Before AD, the aeration step can be common practice for the industrial treatment of OFMSW in garage-type reactors since it reduces the excess of organic acids that can negatively impact the anaerobic process.[26] Considering the complex AD process, the stages of hydrolysis and acidogenesis are considered the limiting stages of methanogenesis, and pretreatment is crucial to increase the methane yield.[27]

The distinction is not clear in the literature between what short and long pre-treatment times would be conducted prior to the AD of FW. Therefore, this study evaluated the effect of two pretreatments on food waste: storage in aerobic conditions and forced aeration prior to AD. The biochemical potential of methane (BMP) was used to analyze the influence of storage time and aeration flow on the anaerobic digestion performance.

Material and methods

Substrate and inoculum

The food waste samples were collected at the restaurant of the Institute of Technological Research (São Paulo, SP, Brazil). The collected residues were fractionated sufficiently to perform the two degradation experiments. In total, 9 kg of FW were obtained after separating the sample from the leftover dishes through a coarse sieve, with about 0.9 kg of liquid discarded and 11.1 kg of preparation residues, consisting of 6.9 kg of melon rinds and 4.2 kg of cassava husks. The samples of 400 g were prepared with 50% in mass of food preparation residues and 50% of the leftover dishes, the first of which were manually cut into sizes not exceeding 1 inch. The samples were stored at −20°C. The mesophilic inoculum was obtained from an Upflow anaerobic sludge blanket digestion reactor of a poultry slaughterhouse (Dacar Company, Tietê, SP, Brazil).

Study of food waste storage for 4 and 7 days

The samples were thawed and placed in 2 L amber glass bottles and suspended using Nylon fabric to separate the leachate from the solid phase. In this experiment, the pretreatment condition was static storage. The 2 L amber

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glass bottles containing the suspended substrates were kept under ambient conditions in a styrofoam box to avoid sudden temperature changes during the storage periods of 4 and 7 days. Two openings at the bottom of the sides and one at the top of the box were provided for natural ventilation. At the end of the storage period, the material suspended in the fabric was weighed, and the volume of the leachate was collected and measured. The material suspended on the Nylon fabric was then mixed with the leachate and homogenized in a blender. The homogenized samples were submitted for analysis of pH, total solid (TS), and total volatile solid (TVS) according to the American Public Health Association (APHA).[28] After the storage steps (0, 4 and 7 days), homogenized samples for each storage period (4 and 7 days) and for the control (0 days) were subjected to the BMP test, which was monitored for 41 days under mesophilic conditions (37°C).

Study of the continuous aeration of food waste for 4 days

A new experiment, similar to the previous one, was conducted to evaluate the influence of continuous aeration for 4 days on the samples. The samples were thawed and placed in 2 L amber glass bottles and suspended in Nylon fabric in the same way as in the previous experiment. The bottles were also placed in a Styrofoam box, but in this experiment, a dry compressed air flux of 1.7 L min^{-1} (equivalent to 36 L air g^{-1} TS day^{-1}) was forced through the solid samples suspended in the bottles by injecting it below the Nylon fabric through a hose. The continuous aeration was maintained for 4 days, with a total aeration of 144 L air g^{-1} TS. At the end of the aeration period, an apparent dryness of the material was observed. The suspended solid material was weighed, and the volume of the leachate in the 2 L bottles was measured. The leachate and the suspended material in the fabric were treated independently. For the homogenization of the suspended material, the content of the triplicate samples was collected, and 100 mL of distilled water was added for processing in a blender. Since the leachate was also partially dried, 100 mL of distilled water was added to each bottle of the triplicate sample to solubilize the material adhering to the bottles. Triplicate samples of the homogenized suspended material and leachate were subjected to analysis of pH, TS and TVS.[28] After the forced aeration, the AD experiment was monitored under mesophilic conditions (37°C), using the same inoculum as used in the previous test.

The elemental analysis of the substrates used in the study was carried out in triplicate using the Flash EA 1112 (Thermo Scientific) equipment. The samples analyzed were microcrystalline cellulose (used as a reference substrate for the BMP tests)[29] and the substrates after 0 and 4 days (suspended solids and leachate). Knowledge of the substrate's chemical composition is essential for calculating the quantity and composition of the biogas produced.[30] Equations (1) and (2) present the Buswell model to estimate the theoretical methane production from the elemental composition:[31]

$$C_nH_aO_bN_c + \left(n - \frac{a}{4} - \frac{b}{2} + \frac{3c}{4}\right)H_2O \rightarrow \left(\frac{n}{2} + \frac{a}{8} - \frac{b}{4} - \frac{3c}{8}\right)CH_4 + \left(\frac{n}{2} - \frac{a}{8} + \frac{b}{4} + \frac{3c}{8}\right)cNH_3 \quad (1)$$

$$TMY\left(mL\,CH_4\,g^{-1}\,TVS\right) = \frac{22.4 \times 1000 \times \left(\frac{n}{2} + \frac{a}{8} - \frac{b}{4} - \frac{3c}{8}\right)}{12n + a + 16b + 14c} \quad (2)$$

where TMY stands for the maximum theoretical methane production converted from the organic matter in the substrate, and *n*, *a*, *b* and *c* refer to the contents of C, H, O, and N atoms, respectively, in the molecular formula of the FW.

Biochemical methane potential

The BMP was performed on the Automatic Methane Potential Test System – AMPTS II (Bioprocess®) equipment according to the conditions of VDI 4630.[32] The amount of inoculum in each reactor represented 1.5% (w/v) of the total reactor volume (500 mL), based on the TVS content. The substrate to inoculum ratio used was 0.5 based on the TVS content, and the resulting total solids in the reactor were 5.2%. The useful volume of the reactor was 375 mL (75%). All of the tests were performed in triplicate. For the BMP of substrates of the storage study, the reactors were prepared with 220 g of inoculum, 153 g of distilled water and 19 g of the substrate (4 and 7 days of storage) or 24 g for the fresh substrate (control reactor). For the reference substrate, 4 g of microcrystalline cellulose was used.[29] For the BMP of the aeration study, the reactors were prepared with 360 g of inoculum (the same inoculum used in the previous BMP test) and 70 mL for the leachate or 14 g for the suspended solid.

Results and discussion

BMP of food waste stored for 4 and 7 days

Table 1 shows the physicochemical characterization of the inoculum and the substrate at the beginning of the experiment and after storage under aerobic conditions. The

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Table 1. Physicochemical characterization of food waste after aerobic storage, inoculum and food waste with leachate.

Parameters	Raw material	Storage time (days)		
		0	4	7
Density ($g\,mL^{-1}$)	Inoculum	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01
pH	Inoculum	7.41 ± 0.01	—	—
	Food waste	3.97 ± 0.09	—	—
TS ($g\,kg^{-1}$)	Inoculum	39.72 ± 4.65	—	—
	Food waste	169.50 ± 3.81	—	—
	Food waste + leachate	—	206.45 ± 2.64	206.09 ± 0.55
TVS ($g\,kg^{-1}$)	Inoculum	33.94 ± 4.05	—	—
	Food waste	158.05 ± 3.79	—	—
	Food waste + leachate	—	197.39 ± 2.44	196.48 ± 0.48
TFS ($g\,kg^{-1}$)	Inoculum	5.78 ± 0.63	—	—
	Food waste	11.45 ± 1.54	—	—
	Food waste + leachate	—	9.06 ± 1.48	9.61 ± 0.55

TFS, total fixed solids; TS, Total solids; TVS, total volatile solids.

results indicate that the fresh samples (0 days of storage) presented lower values of TS when compared with the treated residues, which can be explained by the evaporation and degradation during the storage period. The experiment did not show a major change in the composition of the residues in the characterization parameters from 4 days of storage to 7 days of storage. At the end of the 4 days, a leachate amount of 76 ± 5 mL was obtained, which represents 18.8%, considering the average mass of 403 ± 15 g used. Additionally, at the end of the 7 days, an amount of leachate of 89 ± 9 mL was obtained, representing 22.4%, of the average mass sample of 394 ± 3 g. The generation of leachate for 4 and 7 days was similar, although there was an increase of around 15% from 4 days to 7 days. Degueurce *et al.*[33] evaluated a storage pretreatment for OFMSW and observed that the conversion from solid to leachate ranged from 15 to 27% for different substrates after 2 weeks of storage, and the largest degraded portion (about 15%) occurred in the first 3 days. In addition, during a period of 5–8 days of storage and considering substrates of different compositions, the percentage of solid matter converted to leachate was 17.5%,[34] corroborating the present study.

The results of the TS and TVS of the inoculum and substrates were used as a basis for the BMP tests. Figure 1(a) shows the daily methane yield obtained for the fresh food waste (control) and after storage periods of 4 and 7 days. The maximum daily methane production for the substrates with different storage periods was obtained on the first day of the experiment. For the microcrystalline cellulose, used as a reference substrate, the maximum production occurred on the third and fifth days. Considering the first day, the fresh samples (FW/0 days) presented higher methane production, followed by the samples stored for 4 and 7 days. The results corroborate other studies, which demonstrated that the pretreatment period with aeration, either passively or through forced aeration, reduces the impact of a high initial load on AD, promoting more stable reactors.[24] The average net cumulative specific methane production of food waste stored for 0, 4, and 7 days and the reference substrate (Fig. 1b) showed that a 7 day pretreatment caused an increase in the cumulative specific methane production.

The BMP tests were considered up to the point when the net methane production started to decrease (when the methane production of the inoculum alone was higher than the production of the substrate). For the microcrystalline cellulose, a BMP yield of 335 NmL $CH_4\,g^{-1}$ TVS was obtained, representing more than 80% of the theoretical value (415 NmL $CH_4\,g^{-1}$ TVS). The yield obtained from the fresh food waste was 375.0 NmL $CH_4\,g^{-1}$ TVS, while for the residue stored for 4 days it was 365.0 NmL $CH_4\,g^{-1}$ TVS and for the residue stored for 7 days it was 425.0 NmL $CH_4\,g^{-1}$ TVS. Thus, the accumulated production up to this point shows that the residue stored for 7 days presented a production 16% higher than that for the residue stored for 4 days. These results may indicate that the extension of the storage period contributed to the consumption of readily available acids, which can cause inhibition and also modify the structure of the material, thus contributing to a more stable process.

The BMP test of food waste stored for 4 days was lower than that of the fresh residue, while the residue stored for 7 days presented the maximum methane potential, although it took twice as long to reach that. This aspect must be considered for viability analyses of plants since, although the yield was

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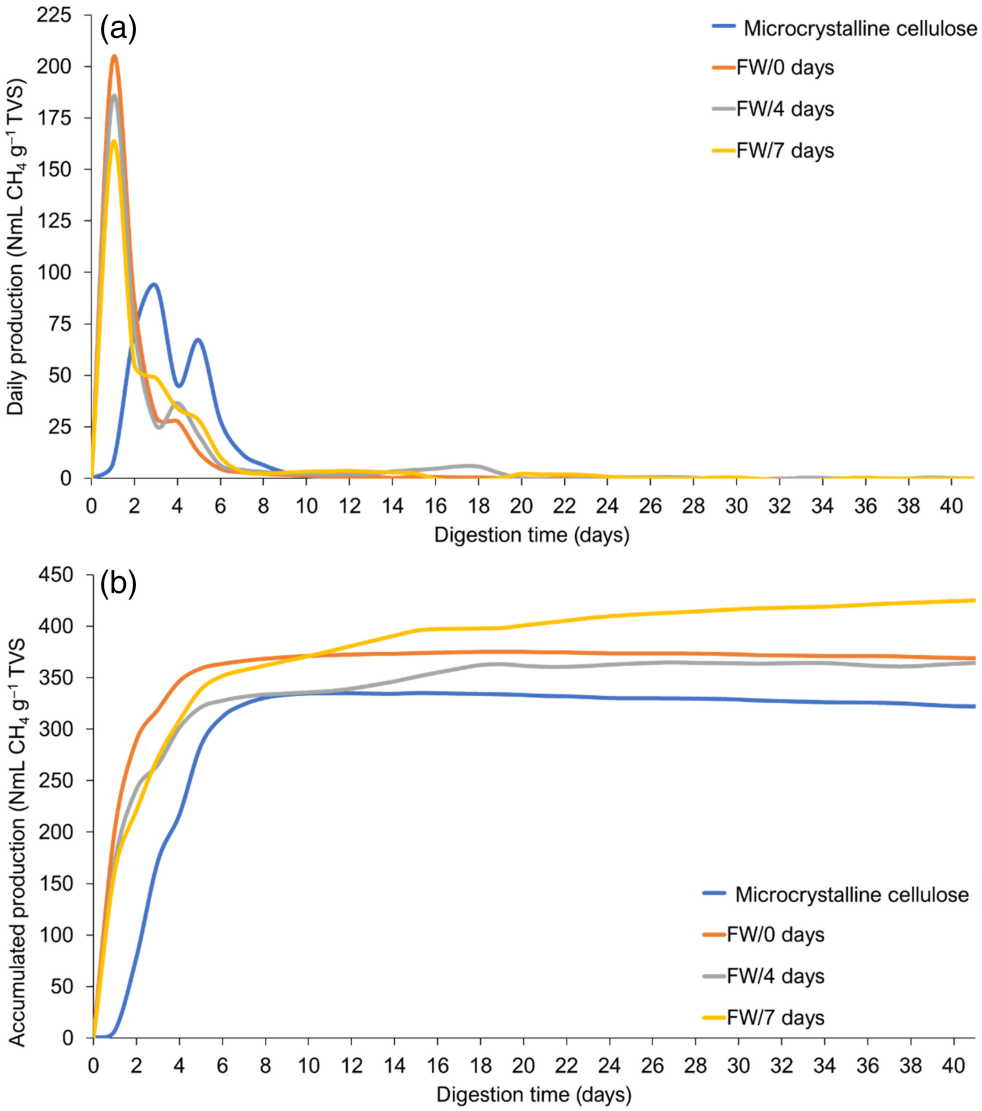


Figure 1. Daily and accumulated methane production obtained for the biochemical potential of methane (BMP) tests of microcrystalline cellulose (reference substrate), and fresh food waste (control FW/0 days) and food waste stored for 4 and 7 days.

better with a longer storage time, the time spent to obtain this production was relevantly longer. Lu *et al.*[35] performed a storage pretreatment under anoxic conditions, concluding that the best results in terms of stability were found with pretreatement periods of 5 and 7 days, with the methane yield ranging from 397 to 484 NmL CH_4 g^{-1} TVS_{added}. This fact indicated that a storage time of around 1 week is an appropriate pretreatment before AD.[33]

BMP of food waste continuously aerated for 4 days

At the end of the 4 day aerating period, the leachate collected presented a volume of 20.5 ± 4.4 mL, approximately 5.3% of the average initial mass (388.2 ± 6.1 g). After continuous aeration for 4 days, 31.8% of the initial mass was lost by evaporation, giving the materials a dry appearance. To avoid this detrimental effect of reduced substrate moisture, which can impair metabolic activities, one possibility would be to humidify the air before aeration. Table 2 presents a characterization of the materials before and after aeration. The residue after aeration was more concentrated owing to evaporation and carbon loss, both in the solid and the liquid fractions. The mass balance after continuous aeration of food waste is presented in Table 3. It was observed that the pretreatment reduced the initial mass by 31.8%, and there was a reduction of 8.5% of TS and 8.3% of TVS. The production reduction was 7.9%.

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Table 2. Physicochemical characterization of inoculum and food waste after continuous aeration for 4 days (suspended material and leachate).

Raw material	Aeration time (days)	TS (g kg^{-1})	TVS (g kg^{-1})	TFS (g kg^{-1})
Inoculum	Day 0	25.09 ± 0.01	20.78 ± 0.01	4.31 ± 0.23
Food waste		210.99 ± 2.26	200.57 ± 2.36	10.42 ± 0.10
Suspended material	Day 4	277.60 ± 1.88	264.50 ± 2.02	13.09 ± 0.43
Leachate		345.23 ± 20.92	312.48 ± 4.90	32.75 ± 2.76

TFS, total fixed solids; TS, total solids; TVS, total volatile solids.

Table 3. The mass balance of solids after 4 days of continuous aeration steps of food waste.

Parameters		Value
Balance based on the total mass		
Initial mass (g)		388.2 ± 6.1
Final mass (g)	Suspended material	244.4 ± 8.1
	Leachate	20.5 ± 4.4
Mass reduction (%)		31.8
Balance based on the TS content		
Initial TS content (g)		81.9 ± 1.3
Final TS content (g)	Suspended material	67.8 ± 2.2
	Leachate	7.1 ± 1.5
TS reduction (%)		8.5
Balance based on the TVS content		
Initial TVS content (g)		77.9 ± 1.2
Final TVS content (g)	Suspended material	65.0 ± 2.1
	Leachate	6.4 ± 1.4
TVS reduction		8.3

TS, total solids; TVS, total volatile solids.

Figure 2(a) shows the specific daily methane production rate in the BMP test of fresh FW, the reference substrate, suspended material and leachate after a continuous aeration period of 4 days. In general, the fresh FW presents a relevantly higher peak on day 1, probably owing to the availability of short-chain acids readily available for consumption. The reference substrate presented two defined peaks, associated with the possible presence and action of hydrognotrophic and later acetoclastic archea.[25,36] The suspended material showed peak production on the first day and a subsequent decrease. On the fifth day, an increase in the activity presented the second-highest production rate. The leachate also presented a peak on the first day, followed by relative inactivity until the fifth day, when methane production restarted gradually to reach a second peak after 11 days of digestion. This fact may be due to the high load of dissolved volatile fatty acids. According to Charles *et al.*,[36] the aerating experiments also showed two peaks in the methane production profile, the first peak referring to the gas produced by hydrogenotrophic methanogens. The second peak was attributed to acetoclastic methenogenics that provided a rapid acetate consumption. This consideration was evidenced by isotope screening, which made it possible to verify that during the first 2 days of the process, there was no conversion from acetate to methane but rather an accumulation of acetate.[36]

From the accumulated methane production (Fig. 2b), it was observed that pretreatment caused an increase in the net cumulative specific methane production. The methane production accumulated in the control reactor increased until the 14th day of digestion, and the highest production speed occurred in up to 5 days for fresh food waste. As expected, a large methane generation occurred on the first day, probably owing to the readily available acids. However, this availability of volatile fatty acids for consumption, especially in larger-scale operating systems, where the inoculum to substrate ratio is not so high as in BMP tests, may cause acidification of the medium, with a great possibility of overload and risks to system collapse.[14] Beyond this, Fig. 2 also shows that the aeration caused an initial reduction of accumulative methane (reduction of the first peak). These results can be explained by the presence of readily available compounds that were consumed in the aerobic process. The suspended material, besides presenting a slow increase in production rate at the beginning, also produced for fewer days when compared with the leachate, so one possibility is that the pretreatment has facilitated hydrolysis, and thus the easily degradable solid materials have been transferred to the liquid fraction, leaving only the most complex degradation materials.

According to Xu *et al.*,[25] a microaeration rate corresponding to 258 L air kg^{-1} TS day^{-1} produced 0.27 L CH_4 g^{-1} TVS_{added}. The FW was aerated at the rate of 387 L air kg^{-1} TS day^{-1} and produced 0.4 L CH_4 g^{-1} TVS_{added}, where the residue without aeration produced 0.2 L CH_4 g^{-1} TVS_{added}. Despite the reported advantages of microaeration, residual oxygen in the acidogenic leach bed reactor can inhibit the activity of methane-producing bacteria and decrease the methane yield.

The leachate and the suspended material showed a reduction in the initial production rate, possibly related to the

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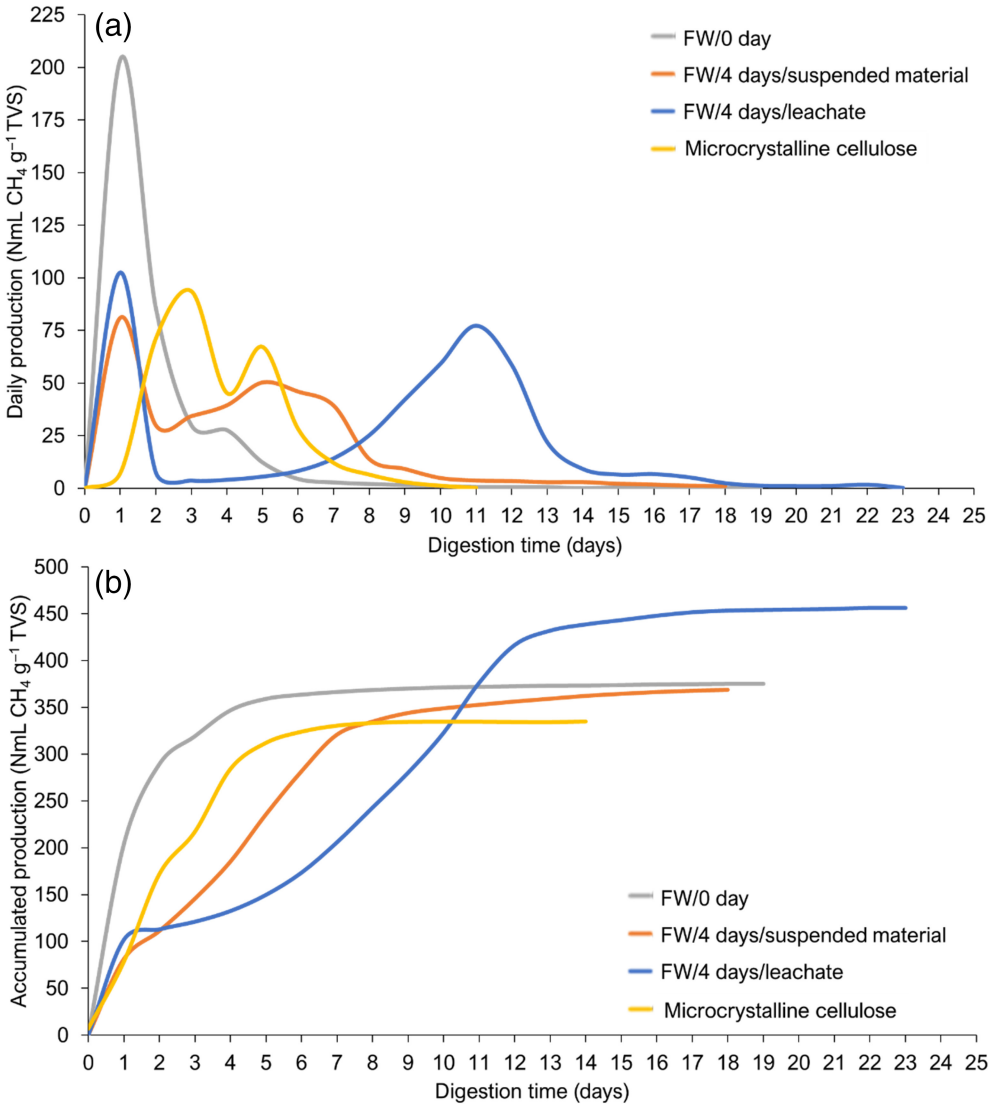


Figure 2. Daily and accumulated methane production obtained for the BMP tests of microcrystalline cellulose (reference substrate), fresh food waste, suspended material and leachate continuously aerated for 4 days.

oxidation of readily available materials. However, the leachate showed a longer production than the solid fraction, reaching the second peak after 11 days (Fig. 2a). These results can be explained by the possibility that the aeration stage provided structural changes in some soluble compounds of more complex degradation, making it possible that after 11 days of AD they could be consumed in the process and more quickly.

The present study compared the experimental with the theoretical methane yield, calculated based on the elemental analysis. Table 4 shows the elemental analysis and ash content of the substrates used in the BMP test after continuous aeration pretreatement. In general, the C/N ratios were 102 (microcrystalline cellulose), 14 (fresh food waste), 16 (4-day aerated suspended material) and 14 (4-day aerated leachate). According to Kondusamy and Kalamdhad,[30] carbon acts as the energy source for microorganisms, and nitrogen plays a role in increasing the microbiota. The recommended C/N ratio for a continuous process is around 25–30, considering the need to meet the nutritional requirements of microorganisms compatible with their feeding rates. In the case of the batch BMP test, this is not necessarily true because there is an excess of microorganisms. Low C/N ratios may represent protein-rich substrates like meats, for example, in the case of food waste.[34] The excess nitrogen in continuous reactors can be toxic to the microorganisms and inhibit the methanogenesis but in batch experiments of the BMP, where the substrate to inoculum ratio is low (< 0.5 in terms of TVS), this imbalance represents little danger of inhibition.

Table 5 presents the molecular formula of each feedstock used to calculate the theoretical methane yield of the

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Table 4. Elemental analysis of the substrates used in the biochemical potential of methane (BMP) of the aerated pretreatement, fresh food waste (FW/0 day), reference substrate (microcrystalline cellulose), suspended material (FW/4 days/suspended materia) and leachate (FW/4 days/leachate) after continuous aeration for 4 days.

Parameters	Composition (%)					C/N
	Nitrogen	Carbon	Hydrogen	Ash	Oxygen	
Microcrystalline cellulose	0.42 ± 0.04	42.85 ± 0.11	6.24 ± 0.05	n.d.	48.84 ± 0.45	102
FW/0 day	3.14 ± 0.07	44.19 ± 0.24	6.51 ± 0.06	10.42 ± 0.41	33.94 ± 0.36	14
FW/4 days/suspended material	2.91 ± 0.15	46.68 ± 0.17	4.64 ± 0.15	11.69 ± 0.23	32.00 ± 0.12	16
FW/4 days/leachate	2.82 ± 0.04	39.82 ± 0.03	6.18 ± 0.11	5.58 ± 0.13	44.00 ± 0.81	14

n.d., Not detected.

Table 5. Molecular formula after the elemetal analyses of the analyzed substrates.

Parameters	Molecular formula
Theoretical cellulose	$C_6H_2O_5 + H_2O \rightarrow 3CH_4 + 3CO_2$
Microcrystalline cellulose	$C_{3.57}H_{6.24}O_{3.05}N_{0.03} + 0.51H_2O \rightarrow 1.79CH_4 + 1.78CO_2 + 0.03NH_3$
FW/0 day	$C_{3.68}H_{6.51}O_{2.12}N_{0.22} + 1.16H_2O \rightarrow 2.04CH_4 + 1.64CO_2 + 0.22NH_3$
FW/4 days/suspended material	$C_{3.89}H_{4.64}O_{2.00}N_{0.21} + 1.89H_2O \rightarrow 1.95CH_4 + 1.94CO_2 + 0.21NH_3$
FW/4 days/leachate	$C_{3.32}H_{6.18}O_{2.75}N_{0.20} + 0.55H_2O \rightarrow 1.67CH_4 + 1.65CO_2 + 0.20NH_3$

Table 6. Efficiency of the process after the biochemical potential of methane (BPM) test in forced aeration for 4 days.

Parameters	TMY (NmL CH_4 g^{-1} TVS)	EMY (NmL CH_4 g^{-1} TVS)	Stabilization time (days)	Difference (%)
Microcrystalline cellulose	414.81 ± 0.42	335.12 ± 2.32	14	19.21
FW/0 day	529.18 ± 0.53	375.34 ± 2.51	19	36.63
FW/4 days/suspended material	512.96 ± 0.86	368.87 ± 1.89	18	28.08
FW/4 days/leachate	430.75 ± 0.32	456.20 ± 3.87	23	5.57

EMY, experimental methane production; TMY, theoretical methane production.

substrates. Table 6 shows the efficiency of the process of the BMP test after continuous aeration pretreatment. The results show a difference of 19.21% for estimating the theoretical BMP for cellulose and that obtained by elemental analysis. Although the leachate reached a theoretical production level lower than that of the solid material, the production efficiency was higher than that theoretically evaluated by the elemental analysis.

Concluding remarks and perspectives

According to the BMP results, a short period of aerobic storage (up to 7 days) and high aeration rates of food waste did not result in a loss of methane-producing potential. Still, it reduced the initial peak of methane production on the first day, which may cause a reletive reduction in the risk of inhibition in the start-up of batch reactors. The results showed that the storage and aeration production reached similar results. The storage took twice as long. This fact, considering the process aspect, is not interesting. The intense and continuous forced aeration applied for 4 days caused an 8.3% reduction in TVS, which may reduce the methane-producing potential. Still, it was compensated for by the higher degradability of both the suspended solid and the leachate. The reduction of the initial methane production peak and the extension of the production period are very important for avoiding methanogenic inhibition. Concerning aerobic storage, the longer the storage period is (up to 7 days), the greater is the attenuation of the initial methane production peak, which can be explained by the oxidation of the readily available substrates. Despite this initial reduction of methane production rate, the accumulated specific production is higher, although over a longer period. This fact has to be taken into account in the design of batch reactors, considering that to achieve the same accumulated production after storage, a longer digestion period will be necessary.

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The proportions of residues for metabolization in AD processes must be evaluated and monitored. Standard conditions are only a reference for industrial applications since the amounts involved in substrates are much lower than the amount of inoculum when solid waste is used. Thus, treatment units should predict lower losses and not operate at maximum production conditions. Although the storage period has contributed to the lowering of the easily degradable organic load, as observed by the production results of the first day, the aeration is more effective. Another highlight is that although the storage (7 days) reached a production level close to that of the aerated substrate for 4 days, it took longer. Therefore, the aeration process is more interesting because it provides faster production, even when the aeration is carried out in excess, as in this case, causing losses of TVS.

Conclusion

This study evaluated the influence of aerobic storage time and forced continuous aeration pretreatment on the BMP of food waste. The results presented suggest that aerobic pretreatment can be used prior to AD to improve biodegradability of food waste and to prevent the inhibition of methanogenesis. For instance, the methane production of the FW stored for 7 days (425 NmL CH_4 g^{-1} TVS) increased when compared with the fresh samples without pretreatment (375 NmL CH_4 g^{-1} TVS). The adoption of forced and continuous aeration pretreatment of food waste for 4 days produced 456.20 NmL CH_4 g^{-1} TVS for leachate, 1.22-fold higher than that for the food waste without storage. Specifically, in large-scale garage-type systems, introducing aeration as a pretreatment can promote the reduction of the excess available organic load in the start-up of the reactors, which otherwise can acidify the reactor owing to the high substrate to inoculum ratio and negative impact on methanogenesis. In conclusion, the application of aeration pretreatment prior to anaerobic digestion can be an alternative to increase the methane potential from food waste.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was supported by the Brazilian Science and Research Foundation (productivity grant 302473/2019-0) and the São Paulo Research Foundation (Brazil) (grant numbers 2018/14938-4 and 2019/26925-7).

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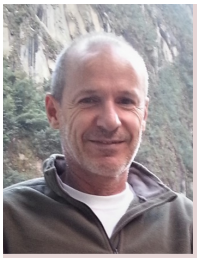
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